PIC 10A 1A

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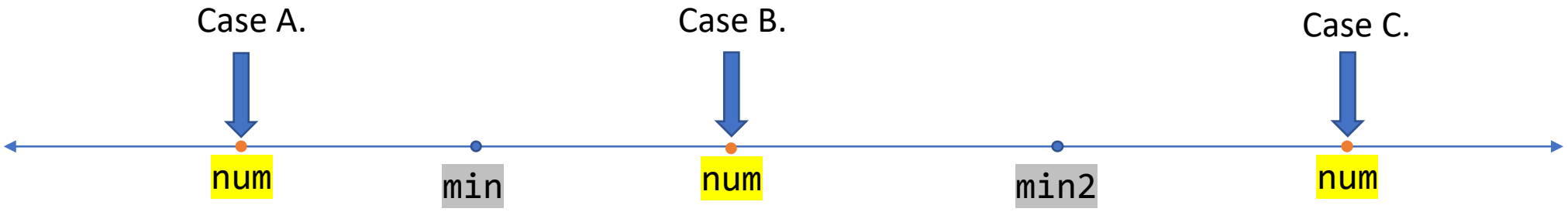
Today…

- HW3 Hints
- Functions

HW3 Hints

HW3 Hints

- Exercise 2: Find the 2nd smallest number in the given stream
  - Keep track of the smallest, and the 2nd smallest (say, `min` and `min2`, respectively)
  - When a new input `num` is given, there are three cases:



  - Case C is the easiest (Do we need to update something?)
  - In Case B, which variable(s) should we update? Does something stay the same in this case?
  - How about Case A?

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HW3 Hints

- If you figured out how to handle those three cases, let's think about how to handle the "same number" cases

- Case A' : num == min
- Case B' : num == min2

- What do we have to do in these cases?

HW3 Hints

- To sum up, the pseudocode may look like:

```
while ( /* input is valid */ ) {
    if ( /* case A */) {
        // do something
        // handle case A' if needed
    }
    else if (/* case B */) {
        // do something
        // handle case A' and/or B' if needed
    }
    else { // case C
        // do anything you need
    }
}
```

- And finally, give some *special care* for “single input” case (e.g. the last ex. in the HW instr.)

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Functions

Functions

- Functions can be
  - Declared and then defined later
    - If you declare the function first (not define it right now) then need ;
  - Defined when it is declared (so both are done at the same time)

- The declaration determines a “signature” of the function
  - [Return Type] [Function Name] [Input Parameters] (and other options later)

```
double find_max(double a, double b);
```

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Functions

- The declaration determines a “signature” of the function
  - [Return Type] [Function Name] [Input Parameters] (and other options later)

```
double find_max(double a, double b);
```

- Return Type
  - The type of the expression returned by the function
  - If the function returns nothing, can be `void`
- Function Name
  - Name of the function; cannot be a reserved word, the same naming rule applies with variables
- Input Parameters
  - The objects passed to the function
  - Number of arguments can be 0, 1, or more
  - Still need parentheses () even when the function gets zero parameters

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Functions – Examples

- A function returning the maximum of two numbers

```
double find_max(double a, double b) {
        return (a > b) ? a : b;
}
```

Recall: The ternary operator

```
A ? B : C
```

is (almost) equivalent to

```
if (A) { B; }
else {C;}
```

- A ***procedure*** is a function returning `void`
  - Which means, it doesn't return anything
  - Example: a function printing the max to the console

```
void print_max(double a, double b) {
    cout << "The max of " << a << " and " << b
         << " is " << find_max(a, b);
}
```

Possible if `find_max` is ***declared*** before this expression

Q: [`true` or `false`?]
Using "`return`" keyword in a ***procedure*** results in a syntax error.

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Functions – Examples

- A function that inputs several different types of arguments

```
void func(unsigned int i, string str) {
        cout << str[i];
}
```

Here `i` cannot be negative

- A ***predicate*** function is a function returning `bool`
  - The following function checks whether the first letter of the string is capitalized or not

```
bool isCapitalized(string str) {
    if ('A' <= str[0] && str[0] <= 'Z') {
        return true;
    }else {
        return false;
    }
}
```

True if `str[0]` is between 'A' and 'Z' (otherwise it is not a capital letter)

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Exercise – Collatz Conjecture

- Given a positive integer, perform the following operation each step:
  - If the number is even, divide it by two $x \rightarrow x/2$
  - If the number is odd, triple it and add one $x \rightarrow 3x + 1$
- The Collatz conjecture states that you will always reach 1.
- Write a program that computes how many steps it takes to reach 1 starting from a positive integer. The input and output should be:

```
Starting from X.
Step 1: X
Step 2: X
...
Step X: X
Starting from X, it took X steps.
```

Output example starting from 6

```
Starting from 6
Step 1: 3
Step 2: 10
Step 3: 5
Step 4: 16
Step 5: 8
Step 6: 4
Step 7: 2
Step 8: 1
Starting from 6, it took 8 Steps.
```

- Hints
  - Again, the `while` loop will be useful
  - This function is a ***procedure***, i.e., it will return `void`

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Your Feedback is welcome

- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)

Click this link

At this point, you may want to request
“more hw hints”,
“more challenging examples/exercises”,
“more details on the classroom examples”, etc.
Please let me know what you want!